STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY & STATISTICS

LECTURE # 21

STATISTICAL INFERENCE
ESTIMATION THEORY

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ESTIMATION THEORY

The purpose of statistics is to use the information contained in a sample to make inference about the population from which the population the sample taken. Populations are characterized by numerical descriptive measures called parameters, the objective of statistical investigations is to estimate the value of one or more relevant parameters.

STATISTICAL INFERENCE The process of drawing inferences about a population on the basis of information contained in a sample taken from the population is called statistical inference. The methods used for statistical inference are

1. Classical Methods
2. Bayesian Methods

In classical methods inferences are based strictly on information obtained from random sample selected from the population while Bayesian methods utilizes the prior subjective knowledge about the probability distribution of the unknown parameters in conjunction with then information provided by the sample data.

Statistical Inference can be divided into two major areas.

1. Estimation
2. Hypothesis Testing

ESTIMATION The procedure of making judgment about a population parameter on the basis of sample is referred to as estimation or statistical estimation. In other words Estimation is a procedure by which we obtain an estimate of the true but unknown value of a population parameter by using the sample observations from the population.

HYPOTHESIS TESTING It is a procedure which enables us to decide about on the basis of information obtained by sampling whether to accept or reject any specified statement or hypothesis regarding the value of the parameter in statistical problem.

ESTIMATOR An estimator is a rule, often expressed as formula that tells how to calculate the value of an estimate based on the measurements contained in a sample.
In other words, *A statistic used to estimate a parameter is called an estimator*. Note that

1. An estimator is also called a decision function.
2. An estimator is random variable because of randomness in selecting the sample.

TYPES OF ESTIMATION Estimation can be divided into two types

1. Point Estimation
2. Interval Estimation

As we know that an actual numerical value obtained from an estimator is called an estimate. When a single number is presented as an estimate of the population parameter, it is known as point estimation, when a range of numbers is provided it is known as interval estimation.

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METHODS OF POINT ESTIMATION

As discussed earlier, when a single number is presented as an estimate of the population parameter, the procedure is known as point estimation. There are two important methods to generate point estimators

1. Method of Moments
2. Method of Maximum Likelihood

We shall discuss Method of Maximum Likelihood only.

METHOD OF MAXIMUM LIKELIHOOD

One of the best methods of obtaining a point estimator of a parameter is the method of maximum likelihood. This technique was developed in the 1920s by a famous British statistician, Sir R. A. Fisher. As the name implies, the estimator will be the value of the parameter that maximizes the **likelihood function**.

> Suppose that X is a random variable with probability distribution $f(x;\theta)$ where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is
>
> $$L(\theta) = f(x_1;\theta)\cdot f(x_2;\theta)\cdot \cdots \cdot f(x_n;\theta) \qquad (7\text{-}10)$$
>
> Note that the likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** (MLE) of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

The estimator produced using this method is known as maximum likelihood estimator (MLE).

PROCEDURE OF FINGING MLE It involves three steps

STEP-1 LIKELIHOOD FUNCTION In this step, we find joint probability function for the given probability mass function or probability density function, we call this likelihood function and it is given by

$$L = f(x_1).\ f(x_2).\ldots.\ f(x_n) \text{----------} (1)$$

simplify the above expression.

STEP-2 LOG LIKELIHOOD FUNCTION In this step, take natural log on both sides of equation (1) and simplify

$$\ln(L) = \ln[\, f(x_1).\ f(x_2).\ldots.\ f(x_n)]$$

$$or \quad l = \ln[\, f(x_1).\ f(x_2).\ldots.\ f(x_n)] \text{------} (2)$$

STEP-3 DERIVATIVES In this step, take derivatives of log likelihood function w.r.t parameters in the probability distribution and put them equals to zero.
Note that Number of derivatives = Number of Parameters, after taking derivatives solve the system of equations and get the MLEs.

EXAMPLES OF MLE

EXAMPLE-1 Let $x_1, x_2, \ldots, x_n$ be a random sample of size 'n' taken from the Bernoulli distribution.

$$f(x;p) = f(x) = p^x q^{1-x}, \quad x = 0, 1, \quad 0 \le p \le 1$$

Find the MLE of p.

SOLUTION The likelihood function for a sample of size 'n' with values $x_1, x_2, \ldots x_n$ is

$$L(p) = f(x_1;p) \cdot f(x_2;p) \cdots f(x_n;p)$$

$$= (p^{x_1}(1-p)^{1-x_1})(p^{x_2}(1-p)^{1-x_2}) \cdots (p^{x_n}(1-p)^{(1-x_n)})$$

$$= p^{x_1+x_2+\cdots+x_n}(1-p)^{n-(x_1+x_2+\cdots+x_n)}$$

$$L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i}$$

Taking ln on BS

$$l = \ln L(p) = \ln\left(p^{\sum x_i}(1-p)^{n-\sum x_i}\right)$$

$$l = \sum x \ln p + (n - \sum x)\ln(1-p)$$

Now put $\frac{\partial l}{\partial p} = 0$

$$\Rightarrow \quad \sum x \cdot \frac{1}{p} \cdot 1 + (n - \sum x) \cdot \frac{1}{(1-p)}(-1) = 0$$

$$\frac{\sum x}{p} = \frac{n - \sum x}{1-p}$$

$$\sum x - p\sum x = np - p\sum x$$

$$p = \frac{\sum x}{n} = \bar{x} \quad \Rightarrow \quad \hat{p} = \bar{x}$$

which is MLE of p.

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EXAMPLE-2 Let X be a Binomial Random variable with parameters n and P. Find the MLE for P.

SOLUTION Let $X_1, X_2, \ldots, X_n$ be a random sample from the Binomial distribution. Now

$$f(x; n, P) = \binom{n}{x} P^x (1-P)^{n-x}, \quad x = 0, 1, 2, \ldots, n.$$

The likelihood function of the sample is

$$L(P) = f(x_1; n, P) \cdot f(x_2, n, P) \cdots f(x_N, n, P)$$

$$= \binom{n}{x_1} P^{x_1} (1-P)^{n-x_1} \cdot \binom{n}{x_2} P^{x_2} (1-P)^{n-x_2} \cdots \binom{n}{x_N} P^{x_N} (1-P)^{n-x_N}$$

$$= \binom{n}{x_1} \cdot \binom{n}{x_2} \cdots \binom{n}{x_N} P^{x_1 + x_2 + \cdots + x_N} (1-P)^{nN - (x_1 + x_2 + \cdots + x_N)}$$

$$L(P) = \binom{n}{x_1} \cdot \binom{n}{x_2} \cdots \binom{n}{x_N} P^{\Sigma x} (1-P)^{nN - \Sigma x}$$

Taking ln on BS

$$\ln L(P) = l = \ln \left[\binom{n}{x_1} \binom{n}{x_2} \cdots \binom{n}{x_N} \right] + \Sigma x \ln P + (nN - \Sigma x) \ln (1-P)$$

Now put $\frac{\partial l}{\partial P} = 0$

$$0 + \Sigma x \cdot \frac{1}{P} \cdot 1 + (nN - \Sigma x) \cdot \frac{1}{(1-P)} (-1) = 0$$

$$\frac{\Sigma x}{P} = \frac{nN - \Sigma x}{1-P}$$

$$\Sigma x - P \Sigma x = nNP - P \Sigma x$$

$$\Sigma x = nNP$$

$$P = \frac{\Sigma x}{nN} = \frac{\bar{x}}{n}$$

Hence MLE for P is $\hat{P} = \frac{\bar{x}}{n}$

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EXAMPLE-3 Let $x_1, x_2, \ldots, x_n$ denote a random sample from the Poisson distribution,

$$f(x;\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Find the Maximum likelihood estimator of λ.

SOLUTION The p.d.f of Poisson distribution is given by

$$f(x;\lambda) = \frac{e^{-\lambda}.\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Now the likelihood function for the sample of size 'n' is given by.

$$L(\lambda) = f(x_1;\lambda) \cdot f(x_2;\lambda) \ldots\ldots f(x_n;\lambda)$$

$$= \left(\frac{e^{-\lambda}.\lambda^{x_1}}{x_1!}\right).\left(\frac{e^{-\lambda}.\lambda^{x_2}}{x_2!}\right)\ldots\left(\frac{e^{-\lambda}.\lambda^{x_n}}{x_n!}\right)$$

$$= \frac{e^{-\lambda-\lambda-\ldots-\lambda} \cdot \lambda^{x_1+x_2+\ldots+x_n}}{x_1! \cdot x_2! \ldots\ldots x_n!}$$

$$L(\lambda) = \frac{e^{-n\lambda}(\lambda)^{\Sigma x}}{x_1!, x_2! \ldots x_n!}$$

Taking ln on B.S we get.

$$l = \ln L(\lambda) = -n\lambda + \Sigma x \ln\lambda - \ln(x_1! \cdot x_2! \ldots x_n!)$$

Now $\frac{\partial l}{\partial \lambda} = 0 \Rightarrow -n + \Sigma x \cdot \frac{1}{\lambda} \cdot 1 - 0 = 0$

$$\Sigma x \cdot \frac{1}{\lambda} = n$$

$$\lambda = \frac{\Sigma x}{n} = \bar{x} \quad \Rightarrow \boxed{\hat{\lambda} = \bar{x}}$$

Hence MLE of Parameter λ of the Poisson distribution is given by $\hat{\lambda} = \bar{x}$.

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EXAMPLE-4 Assume that the random variable X has the exponential distribution.

$$f(x;\theta) = \theta \cdot e^{-\theta x}, \quad x>0, \ \theta>0$$

what is the Maximum Likelihood estimator for θ based on a random sample of n observations of $\bar{x}$.

SOLUTION Let $x_1, x_2, x_3, \ldots, x_n$ be a random sample from the density function.

The likelihood function of the sample is

$$L(\theta) = f(x_1;\theta) \cdot f(x_2;\theta) \cdots f(x_n;\theta)$$

$$= (\theta e^{-\theta x_1}) \cdot (\theta \cdot e^{-\theta x_2}) \cdots (\theta \cdot e^{-\theta x_n})$$

$$= \theta^{1+1+\cdots+1} \cdot e^{-\theta(x_1+x_2+\cdots+x_n)}$$

$$L(\theta) = \theta^n \cdot e^{-\theta \Sigma x}$$

Taking ln on B.S we get.

$$\ln L(\theta) = \ln(\theta^n e^{\theta \Sigma x})$$

$$l = n \ln \theta - \theta \Sigma x$$

Now $$\frac{\partial l}{\partial \theta} = 0 \Rightarrow n \cdot \frac{1}{\theta} \cdot 1 - 1 \cdot \Sigma x = 0$$

$$\frac{n}{\theta} = \Sigma x$$

$$\theta = \frac{n}{\Sigma x} = \frac{1}{\Sigma x/n} = \frac{1}{\bar{x}}$$

Hence MLE of θ is given by $\hat{\theta} = \frac{1}{\bar{x}}$.

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EXAMPLE-⑤ Find MLE of the Mean (μ) and variance (σ^2) of a $N(\mu, \sigma^2)$ based on a Random Sample of size 'n'.

SOLUTION Let $x_1, x_2, \ldots, x_n$ denote a Random Sample of size 'n' from a Normal distribution.

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty$$

The Likelihood function of the Sample is

$$L(\mu, \sigma^2) = f(x_1; \mu, \sigma^2) \cdot f(x_2; \mu, \sigma^2) \cdots f(x_n; \mu, \sigma^2)$$

$$= \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_1-\mu)^2}\right)\left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_2-\mu)^2}\right) \cdots \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_n-\mu)^2}\right)$$

$$= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n e^{-\frac{1}{2\sigma^2}[(x_1-\mu)^2 + (x_2-\mu)^2 + \cdots + (x_n-\mu)^2]}$$

$$L(\mu, \sigma^2) = \left(\frac{1}{\sigma^2 \cdot 2\pi}\right)^{n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

Taking ln on BS we get.

$$l = \ln L(\mu, \sigma^2) = -\frac{n}{2}\ln\sigma^2 - \frac{n}{2}\ln 2\pi - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Now

$$\frac{\partial l}{\partial \mu} = 0$$

$$\Rightarrow \frac{-1}{2\sigma^2} \Sigma 2(x-\mu)^{2-1}(-1) = 0$$

$$\Sigma(x-\mu) = 0$$

$$\Sigma x - \Sigma\mu = 0$$

$$\Rightarrow \Sigma x - \Sigma\mu = 0$$

$$\Sigma x - n\mu = 0$$

$$\Sigma x = n\mu$$

$$\mu = \frac{\Sigma x}{n} = \bar{x}$$

$$\boxed{\hat{\mu} = \bar{x}}$$

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Again $\frac{\partial l}{\partial \sigma^2} = 0$

$$\Rightarrow -\frac{n}{2} \cdot \frac{1}{\sigma^2} \cdot 1 - 0 + \frac{1}{2} (\sigma^2)^{-2} \cdot 1 \Sigma (x-\mu)^2 = 0$$

$$-\frac{n}{2} \cdot \frac{1}{\sigma^2} = -\frac{1}{2} \cdot \frac{1}{\sigma^4} \Sigma (x-\mu)^2$$

$$-\frac{n}{\sigma^2} = -\frac{1}{\sigma^4} \Sigma (x-\mu)^2$$

$$-n = -\frac{1}{\sigma^2} \Sigma (x-\mu)^2$$

$$n = \frac{1}{\sigma^2} \Sigma (x-\mu)^2$$

$$\sigma^2 = \frac{\Sigma (x-\mu)^2}{n}$$

$$\sigma^2 = \frac{\Sigma (x-\bar{x})^2}{n}$$

$$\sigma^2 = s^2$$

Hence the joint ML estimators of μ and σ^2 are the sample mean $\bar{x}$ and sample variance s^2, which is not unbiased estimator of σ^2.

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POINT ESTIMATION

The object of point estimation is to obtain a single number from the sample which will represent the unknown value of the parameter. Population Parameters (such as Population Mean, Population variance etc) are estimated from the corresponding sample statistics (sample mean, sample variance etc).

In general, we use the symbol θ to denote the population parameter which may be proportion, mean or some measure of variability. The parameter θ is unknown constant. The available information is in the form of a random sample $x_1, x_2, \ldots, x_n$ of size n drawn from the population. We wish to formulate a function of the sample observations $x_1, x_2, \ldots, x_n$ that is, we look for a statistic such that its value computed from the sample data would reflect the value of the population parameter as closely as possible. The estimator of θ is commonly denoted by $\hat{\theta}$. Different random samples usually provides different values of the statistic $\hat{\theta}$. Thus $\hat{\theta}$ is a random variable with its own sampling distribution.

PROPERTIES OF A GOOD POINT ESTIMATOR

A point estimator is considered a good estimator if it satisfies various criteria. The four commonly used criterion are as follows

1. unbiasedness
2. consistency
3. Efficiency
4. Sufficiency.

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UNBIASEDNESS

An estimator $\hat{\theta}$ is said to be unbiased estimator of Population Parameter 'θ' if the expected value of the estimator is equal to the true population parameter being estimated. ie if $E(\hat{\theta}) = \theta$.

If $E(\hat{\theta}) \neq \theta$ then $E(\hat{\theta}) = \theta + k$ then 'k' is called 'Bias'.

Thus Bias is defined as 'the difference between the expected value and the true value is known as Bias of the estimator. It is given by

$$\text{Bias} = E(\hat{\theta}) - E(\theta).$$

The estimator is said to be positively biased if $E(\hat{\theta}) > \theta$ and it is said to be negatively biased if $E(\hat{\theta}) < \theta$.

The chief merit of unbiasedness estimator is that it can be easily combined. Thus if $\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_k$ are all the unbiased estimators of θ and are obtained from different sources then

$$\hat{\phi} = \sum_{i=1}^{k} a_i \hat{\theta}_i \qquad \left(\sum_{i=1}^{k} a_i = 1\right)$$

is another estimator of 'θ'.

check: $E(\hat{\phi}) = E\left(\sum_{i=1}^{k} a_i \hat{\theta}_i\right)$

$$= E[a_1\hat{\theta}_1 + a_2\hat{\theta}_2 + \cdots + a_k\hat{\theta}_k]$$

$$= a_1 E(\hat{\theta}_1) + a_2 E(\hat{\theta}_2) + \cdots + a_k E(\hat{\theta}_k)$$

$$= a_1\theta + a_2\theta + \cdots + a_k\theta$$

$$= \theta(a_1 + a_2 + \cdots + a_k)$$

$$= \theta(1) = \theta \qquad \text{as } \sum_{i=1}^{k} a_i = 1$$

$\Rightarrow \hat{\phi}$ is an unbiased estimator of θ.

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EXAMPLES of UNBIASEDNESS

EXAMPLE-1 Show that sample Mean is an unbiased estimator of the population Mean.

SOLUTION Let $\bar{X}$ be the mean of random sample of size 'n' from a population with mean μ. Now we have to show that $E(\bar{X}) = \mu$.

Let $x_1, x_2, \ldots, x_n$ be a random sample of size 'n' from a Population with mean 'μ' then

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Now

$$E(\bar{X}) = E\left(\frac{1}{n}\Sigma X\right)$$

$$= \frac{1}{n} E(\Sigma X)$$

$$= \frac{1}{n} E[x_1 + x_2 + \cdots + x_n]$$

$$E(\bar{X}) = \frac{1}{n}[E(x_1) + E(x_2) + \cdots + E(x_n)]$$

In a random sample, the sample variables $x_1, x_2, \ldots, x_n$ are independent and each has the same distribution of the population. Thus

$$E(x_1) = E(x_2) = \cdots = E(x_n) = \mu$$

Thus $E(\bar{X}) = \frac{1}{n}[\mu + \mu + \cdots + \mu] = \frac{n\mu}{n} = \mu$

$\Rightarrow$ Sample Mean is an unbiased estimator of Pop Mean.

EXAMPLE-2 Show that sample variance defined by $s^2 = \frac{1}{(n-1)}\Sigma(x-\bar{x})^2$ is an unbiased estimator of Population variance σ^2.

SOLUTION We have to show that $E(s^2) = \sigma^2$

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$$LHS = E(s^2)$$

$$= E\left(\frac{1}{(n-1)} \Sigma (x-\bar{x})^2\right)$$

$$= E\left[\frac{1}{(n-1)} \Sigma (x-\mu+\mu-\bar{x})^2\right]$$

$$= \frac{1}{(n-1)} E\left[\Sigma (x-\mu)^2 + \Sigma (\mu-\bar{x})^2 + 2\Sigma (x-\mu)(\mu-\bar{x})\right]$$

$$= \frac{1}{(n-1)} E\left[\Sigma (x-\mu)^2 + n(\bar{x}-\mu)^2 - 2[(\Sigma x - \Sigma\mu)(\bar{x}-\mu)]\right]$$

$$= \frac{1}{(n-1)} E\left[\Sigma (x-\mu)^2 + n(\bar{x}-\mu)^2 - 2(n\bar{x}-n\mu)(\bar{x}-\mu)\right]$$

$$= \frac{1}{(n-1)} E\left[\Sigma (x-\mu)^2 + n(\bar{x}-\mu)^2 - 2n(\bar{x}-\mu)^2\right]$$

$$= \frac{1}{(n-1)} E\left[\Sigma (x-\mu)^2 - n(\bar{x}-\mu)^2\right]$$

$$= \frac{1}{(n-1)} \left[\Sigma E(x-\mu)^2 - n E(\bar{x}-\mu)^2\right]$$

$$= \frac{1}{(n-1)} \left[\Sigma \text{var}(x) - n \text{var}(\bar{x})\right]$$

$$= \frac{1}{(n-1)} \left[n\sigma^2 - n \cdot \frac{\sigma^2}{n}\right]$$

$$= \frac{1}{(n-1)} (n-1)\sigma^2$$

$$E(s^2) = \sigma^2$$

$\Rightarrow$ Sample variance defined by $s^2 = \frac{\Sigma (x-\bar{x})^2}{n-1}$ is an unbiased estimator of population variance σ^2.

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EXAMPLE-(3) Show that if $X \sim B(n,p)$ then the sample proportion $\frac{X}{n}$ is an unbiased estimator of the parameter p.

SOLUTION We have to show that $E(\frac{X}{n}) = p$.

$$LHS = E\left(\frac{X}{n}\right)$$
$$= \frac{1}{n} E(X)$$
$$= \frac{1}{n} \times np \quad (\text{As for Binomial dist. } E(X) = np)$$
$$= p = RHS.$$

$\Rightarrow$ If $X \sim B(n,p)$ then $\frac{X}{n}$ is an unbiased estimator of the Parameter p.

EXAMPLE-(4) A random sample of size n is drawn from the rectangular distribution.

$$f(x) = \frac{1}{\theta}, \quad 0 \le x \le \theta$$

Show that the statistic $2\bar{x}$ is an unbiased estimator of θ.

SOLUTION We have to show that $E(2\bar{X}) = \theta$.

$$LH.S = E(2\bar{X})$$
$$= 2E(\bar{X})$$
$$= 2E(X) \quad [\text{As } E(\bar{X}) = \mu = E(X) \text{ as } \bar{X} \text{ is an unbiased est of } \mu]$$
$$= 2\int_0^\theta x \cdot \frac{1}{\theta}\, dx = \frac{2}{\theta}\int_0^\theta x\, dx$$
$$= \frac{2}{\theta} \cdot \left|\frac{x^2}{2}\right|_0^\theta = \frac{1}{\theta}[\theta^2 - 0]$$
$$= \theta = RHS. \Rightarrow 2\bar{X} \text{ is an unbiased estimator of } \theta.$$

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EFFICIENCY

An unbiased estimator is said to be an efficient estimator if the variance of its sampling distribution is smaller than that of the sampling distribution of any other unbiased estimator of the same parameter.

An estimator is said to be more efficient than another if it has smaller sample variance, in other words suppose there are two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of the same parameter θ, then $\hat{\theta}_1$ will be more efficient estimator than $\hat{\theta}_2$ if $Var(\hat{\theta}_1) < Var(\hat{\theta}_2)$.

$\Rightarrow$ $\hat{\theta}_1$ estimates 'θ' more efficiently than $\hat{\theta}_2$.

Among all unbiased estimators, the one having the smallest variance is refferred to as the Minimum Variance unbiased estimator.

Relative efficiency of an estimator is defined as by the following ratio

$$\text{Relative efficiency} = \frac{Var(\hat{\theta}_1)}{Var(\hat{\theta}_2)}$$

If $\frac{Var(\hat{\theta}_1)}{Var(\hat{\theta}_2)} < 1 \Rightarrow \hat{\theta}_1$ is more efficient than $\hat{\theta}_2$.

MATHEMATICAL CRITERIA

An estimator $\hat{\theta}$ is said to be an efficient estimator iff there exists a function $M(\theta)$ and $V(\theta)$ which may depends upon parameter 'θ' but does not depend upon sample observations: $x_1, x_2, \ldots, x_n$ and which satisfy the following criteria

$$\hat{\theta} = M(\theta).\left\{\frac{\partial}{\partial\theta}.\ln(L(x;\theta))\right\} + V(\theta)$$

where $L(x;\theta)$ is the likelihood function.

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EXAMPLES OF EFFICIENCY

EXAMPLE-1 Let X_1, X_2, X_3 be a random sample of size 3 from population with mean μ and variance σ^2. consider the following estimators of the mean μ.

$$\hat{\theta}_1 = \frac{X_1 + X_2 + X_3}{3} \quad \text{(Sample Mean)}$$

$$\hat{\theta}_2 = \frac{X_1 + 2X_2 + X_3}{4} \quad \text{(Weighted Mean)}$$

which estimator should be preffered?

SOLUTION First we show that $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased then we check efficiency.

$$E(\hat{\theta}_1) = E\left(\frac{X_1 + X_2 + X_3}{3}\right) = \frac{1}{3} E(X_1 + X_2 + X_3)$$

$$// \quad = \frac{1}{3}[E(X_1) + E(X_2) + E(X_3)]$$

$$E(\hat{\theta}_1) = \frac{1}{3}[\mu + \mu + \mu] = \mu \Rightarrow \hat{\theta}_1 \text{ is unbiased estimator of } \mu.$$

$$E(\hat{\theta}_2) = E\left(\frac{X_1 + 2X_2 + X_3}{4}\right)$$

$$// \quad = \frac{1}{4}[E(X_1) + 2E(X_2) + E(X_3)]$$

$$E(\hat{\theta}_2) = \frac{1}{4}[\mu + 2\mu + \mu] = \mu \Rightarrow \hat{\theta}_2 \text{ is unbiased estimator of } \mu.$$

Now

$$Var(\hat{\theta}_1) = Var\left(\frac{X_1 + X_2 + X_3}{3}\right) = \frac{1}{9}[Var(X_1) + Var(X_2) + Var(X_3)]$$

$$= \frac{1}{9}(\sigma^2 + \sigma^2 + \sigma^2) = \frac{1}{3}\sigma^2$$

$$Var(\hat{\theta}_2) = Var\left(\frac{X_1}{4} + \frac{1}{2}X_2 + \frac{1}{4}X_3\right) = \frac{1}{16}Var(X_1) + \frac{1}{4}Var(X_2) + \frac{1}{16}Var(X_3)$$

$$= \frac{1}{16}\sigma^2 + \frac{1}{4}\sigma^2 + \frac{1}{16}\sigma^2 = \frac{3}{8}\sigma^2$$

$$\Rightarrow Var(\hat{\theta}_1) < Var(\hat{\theta}_2)$$

$\Rightarrow$ $\hat{\theta}_1$ is a better estimator than $\hat{\theta}_2$.

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EXAMPLE-2 Let X_1, X_2, X_3 and X_4 be a random sample of size $n=4$ from a $N(\mu, \sigma^2)$. A statistician wishes to estimate the mean by using either of the following two estimators of the mean μ

$$T_1 = \frac{X_1+X_2+X_3+X_4}{4} \quad \text{(Sample Mean)}$$

$$T_2 = \frac{X_1+2X_2+3X_3+X_4}{7} \quad \text{(Weighted Mean)}$$

which estimator should be preferred?

SOLUTION First we show that estimators are unbiased.

$$E(T_1) = E\left(\frac{X_1+X_2+X_3+X_4}{4}\right) = \frac{1}{4}(\mu+\mu+\mu+\mu) = \mu \Rightarrow T_1 \text{ is UBE}$$

$$E(T_2) = E\left(\frac{X_1+2X_2+3X_3+X_4}{7}\right) = \frac{1}{7}(\mu+2\mu+3\mu+\mu) = \mu \Rightarrow T_2 \text{ is UBE.}$$

Now

$$Var(T_1) = Var\left(\frac{X_1+X_2+X_3+X_4}{4}\right)$$

$$= \frac{1}{16}(\sigma^2+\sigma^2+\sigma^2+\sigma^2)$$

$$= \frac{4}{16}\sigma^2 = \frac{1}{4}\sigma^2$$

$$Var(T_2) = Var\left(\frac{1}{7}X_1+\frac{2}{7}X_2+\frac{3}{7}X_3+\frac{1}{7}X_4\right)$$

$$= \frac{1}{49}\sigma^2+\frac{4}{49}\sigma^2+\frac{9}{49}\sigma^2+\frac{1}{49}\sigma^2$$

$$= \frac{15}{49}\sigma^2$$

$\Rightarrow Var(T_1) < Var(T_2)$

$\Rightarrow T_1$ is more efficient than T_2.

QUESTION Let X_1, X_2 be a random sample of size 2 from a $N(\mu, \sigma^2)$. If $\hat{\theta}_1 = \frac{X_1+X_2}{2}$, $\hat{\theta}_2 = \frac{X_1+2X_2}{3}$ which estimator is more efficient?

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MEAN SQUARE ERROR

Sometimes it is necessary to use a biased estimator. In such cases, the mean squared error of the estimator can be important. The **mean squared error** of an estimator $\hat{\theta}$ is the expected squared difference between $\hat{\theta}$ and θ.

DEFINITION The mean square error of an estimator $\hat{\theta}$ is defined as

$$MSE(\hat{\theta}) = E\left(\hat{\theta} - \theta\right)^2$$

Mean Square Error can be written as

$$MSE(\hat{\theta}) = V\left(\hat{\theta}\right) + (Bias)^2$$

That is, the mean squared error of $\hat{\theta}$ is equal to the variance of the estimator plus the squared bias. If $\hat{\theta}$ is an unbiased estimator of θ, the mean squared error of $\hat{\theta}$ is equal to the variance of $\hat{\theta}$.

PROVE THAT $MSE(\hat{\theta}) = V\left(\hat{\theta}\right) + (Bias)^2$

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= E((\hat{\theta} - \theta)^2) \\
&= E(((\hat{\theta} - E(\hat{\theta})) + (E(\hat{\theta}) - \theta))^2) \\
&= E((\hat{\theta} - E(\hat{\theta}))^2) + 2(E(\hat{\theta}) - \theta)E(\hat{\theta} - E(\hat{\theta})) + (E(\hat{\theta}) - \theta)^2
\end{aligned}$$

However,

$$E(\hat{\theta} - E(\hat{\theta})) = E(\hat{\theta}) - E(\hat{\theta}) = 0$$

so that

$$\text{MSE}(\hat{\theta}) = E((\hat{\theta} - E(\hat{\theta}))^2) + (E(\hat{\theta}) - \theta)^2 = \text{Var}(\hat{\theta}) + \text{bias}^2$$

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EXAMPLES OF MEAN SQUARE ERROR

EXAMPLE-1 Suppose that $E(X_1)=\mu, Var(X_1)=10, E(X_2)=\mu$ *and* $Var(X_2)=15$, and consider the point estimates

$$\hat{\mu}_1=\frac{X_1}{2}+\frac{X_2}{2} \quad and \quad \hat{\mu}_2=\frac{X_1}{4}+\frac{3X_2}{4}$$

(a) Calculate the bias of each point estimate. Is any one of them unbiased?
(b) Calculate the variance of each point estimate. Which one has the smallest variance?
(c) Calculate the mean square error of each point estimate. Which one has the smallest MSE?

SOLUTION (a) UNBIASEDNESS

$$E(\hat{\mu}_1)=E\left(\frac{X_1}{2}+\frac{X_2}{2}\right)=\frac{1}{2}E(X_1)+\frac{1}{2}E(X_2)=\frac{1}{2}\mu+\frac{1}{2}\mu=\mu$$

$\Rightarrow \hat{\mu}_1$ *is an unbiased estimator of* μ. *So* $Bias(\hat{\mu}_1)=E(\hat{\mu}_1)-\mu=\mu-\mu=0$

$$E(\hat{\mu}_2)=E\left(\frac{X_1}{4}+\frac{3X_2}{4}\right)=\frac{1}{4}E(X_1)+\frac{3}{4}E(X_2)=\frac{1}{4}\mu+\frac{3}{4}\mu=\mu$$

$\Rightarrow \hat{\mu}_2$ *is an unbiased estimator of* μ. *So* $Bias(\hat{\mu}_2)=E(\hat{\mu}_2)-\mu=\mu-\mu=0$

So both estimators are unbiased.

(b) VARIANCES

$$V(\hat{\mu}_1)=V\left(\frac{X_1}{2}+\frac{X_2}{2}\right)=\frac{1}{4}V(X_1)+\frac{1}{4}V(X_2)=\frac{1}{4}(10)+\frac{1}{4}(15)=6.25$$

$$V(\hat{\mu}_2)=V\left(\frac{X_1}{4}+\frac{3X_2}{4}\right)=\frac{1}{16}V(X_1)+\frac{9}{16}V(X_2)=\frac{1}{16}(10)+\frac{9}{16}(15)=9.0625$$

$\Rightarrow \hat{\mu}_1$ *has smallest* variance *than* $\hat{\mu}_2$.

(c) MEAN SQUARE ERRORS $MSE(\hat{\theta})=V(\hat{\theta})+(Bias)^2$

$MSE(\hat{\mu}_1)=V(\hat{\mu}_1)+(Bias)^2$
Here $V(\hat{\mu}_1)=6.25$ *also* $Bias(\hat{\mu}_1)=0$
So $MSE(\hat{\mu}_1)=V(\hat{\mu}_1)+(Bias)^2=6.25+0^2=6.25$
$MSE(\hat{\mu}_2)=V(\hat{\mu}_2)+(Bias)^2$
Here $V(\hat{\mu}_2)=9.0625$ *also* $Bias(\hat{\mu}_2)=0$
So $MSE(\hat{\mu}_2)=V(\hat{\mu}_2)+(Bias)^2=9.0625+0^2=9.0625$
$\Rightarrow \hat{\mu}_1$ *has smallest MSE than* $\hat{\mu}_2$. *So* $MSE(\hat{\mu}_1)<MSE(\hat{\mu}_2)$

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EXAMPLE-2 Suppose that $E(X_1)=\mu, Var(X_1)=10, E(X_2)=\mu$ *and* $Var(X_2)=15$, and consider the point estimates

$$\hat{\mu}_1=\frac{X_1}{2}+\frac{X_2}{2}, \qquad \hat{\mu}_2=\frac{X_1}{4}+\frac{3X_2}{4} \quad \text{and} \quad \hat{\mu}_3=\frac{X_1}{6}+\frac{X_2}{3}+9$$

(a) Calculate the bias of each point estimate. Is any one of them unbiased?
(b) Calculate the variance of each point estimate. Which one has the smallest variance?
(c) Calculate the mean square error of each point estimate. Which point estimate has the smallest mean square error when $\mu = 8$?

SOLUTION (a) UNBIASEDNESS

$$E(\hat{\mu}_1)=E\left(\frac{X_1}{2}+\frac{X_2}{2}\right)=\frac{1}{2}E(X_1)+\frac{1}{2}E(X_2)=\frac{1}{2}\mu+\frac{1}{2}\mu=\mu$$

$\Rightarrow \hat{\mu}_1$ *is an unbiased estimator of* μ. *So* $Bias(\hat{\mu}_1)=E(\hat{\mu}_1)-\mu=\mu-\mu=0$

$$E(\hat{\mu}_2)=E\left(\frac{X_1}{4}+\frac{3X_2}{4}\right)=\frac{1}{4}E(X_1)+\frac{3}{4}E(X_2)=\frac{1}{4}\mu+\frac{3}{4}\mu=\mu$$

$\Rightarrow \hat{\mu}_2$ *is an unbiased estimator of* μ. *So* $Bias(\hat{\mu}_2)=E(\hat{\mu}_2)-\mu=\mu-\mu=0$

$$E(\hat{\mu}_3)=E\left(\frac{X_1}{6}+\frac{X_2}{3}+9\right)=\frac{1}{6}E(X_1)+\frac{1}{3}E(X_2)+9=\frac{1}{6}\mu+\frac{1}{3}\mu+9=(1/2)\mu+9\neq\mu$$

$\Rightarrow \hat{\mu}_3$ *is a biased estimator of* μ. *So* $Bias(\hat{\mu}_3)=E(\hat{\mu}_3)-\mu=(1/2)\mu+9-\mu=9-(1/2)\mu$

So $\hat{\mu}_1$ *and* $\hat{\mu}_2$ *both estimators are unbiased but* $\hat{\mu}_3$ *is a biased estimator of* μ.

(b) VARIANCES

$$V(\hat{\mu}_1)=V\left(\frac{X_1}{2}+\frac{X_2}{2}\right)=\frac{1}{4}V(X_1)+\frac{1}{4}V(X_2)=\frac{1}{4}(10)+\frac{1}{4}(15)=6.25$$

$$V(\hat{\mu}_2)=V\left(\frac{X_1}{4}+\frac{3X_2}{4}\right)=\frac{1}{16}V(X_1)+\frac{9}{16}V(X_2)=\frac{1}{16}(10)+\frac{9}{16}(15)=9.0625$$

$$V(\hat{\mu}_3)=V\left(\frac{X_1}{6}+\frac{X_2}{3}+9\right)=\frac{1}{36}V(X_1)+\frac{1}{9}V(X_2)+0=\frac{1}{36}(10)+\frac{1}{9}(15)=1.94$$

$\Rightarrow \hat{\mu}_3$ *has smallest* variance.

(c) MEAN SQUARE ERRORS $MSE(\hat{\theta})=V(\hat{\theta})+(Bias)^2$

$MSE(\hat{\mu}_1)=V(\hat{\mu}_1)+(Bias)^2$ *Here* $V(\hat{\mu}_1)=6.25$ *also* $Bias(\hat{\mu}_1)=0$

So $MSE(\hat{\mu}_1)=V(\hat{\mu}_1)+(Bias)^2=6.25+0^2=6.25$

$MSE(\hat{\mu}_2)=V(\hat{\mu}_2)+(Bias)^2$ *Here* $V(\hat{\mu}_2)=9.0625$ *also* $Bias(\hat{\mu}_2)=0$

So $MSE(\hat{\mu}_2)=V(\hat{\mu}_2)+(Bias)^2=9.0625+0^2=9.0625$

$MSE(\hat{\mu}_3)=V(\hat{\mu}_3)+(Bias)^2$ *Here* $V(\hat{\mu}_3)=1.94$ *also* $Bias(\hat{\mu}_3)=9-(1/2)\mu=9-(1/2)(8)=5$

So $MSE(\hat{\mu}_3)=V(\hat{\mu}_3)+(Bias)^2=1.94+(5)^2=26.94$

$\Rightarrow \hat{\mu}_1$ *has smallest MSE*.

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